

! SpringerOpen journals have moved to Springer Nature Link. [Learn more about website changes.](#)

SPRINGER NATURE Link

Log in

Menu

Search

Cart

[Home](#) > [EPJ Data Science](#) > Article

A hybrid stock prediction method based on periodic/non-periodic features analyses

| Research | [Open access](#) | Published: 03 January 2025

| Volume 14, article number 1, (2025) | [Cite this article](#)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Accept all cookies

Reject optional cookies

Manage preferences

Stock investment is an economic activity characterized by high risks and high returns. Therefore, the prediction of stock prices or fluctuations is of great importance to investors. Stock price prediction is a challenging task due to the nonlinearity and high volatility of stock time series. Existing deep learning models may not capture the periodic and non-periodic features of stock data effectively. In this paper, we propose a novel model that leverages Complete Ensemble Empirical Mode Decomposition (CEEMD), Time2Vec, and Transformer to better capture and utilize various patterns in stock data for enhanced prediction performance, and we call it ETT. CEEMD decomposes the stock data into different frequency components based on their intrinsic scales. Time2Vec provides a time vector representation that captures both periodic and non-periodic patterns while being invariant to time scaling. Transformer learns the long-term dependencies and global information from the data. We apply ETT to predict stock prices in the Chinese A-share market and compare it with several baseline models. The results show that ETT reduces the mean squared error (MSE) by an average of 4% and increases the average cumulative return by

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

[Fourier Analysis](#)[Financial Econometrics](#)[Machine Learning](#)[Quantitative Finance](#)[Statistical Finance](#)[Structure Prediction](#)[Financial Time Series Prediction Using Machine Learning Techniques](#)

1 Introduction

The stock market is an important component of a market economy and serves as an economic barometer. With the development of the economy and the improvement of people's investment awareness, an increasing number of individuals are participating in it. However, the fluctuations in stock prices are influenced by various factors such as market conditions, politics, macroeconomics, and investor psychology [1, 2]. These factors result in a mixture of

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [**privacy policy**](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

proposed an innovative deep learning architecture called Transformer [10], which utilizes attention mechanisms to process data and enables parallel training while capturing global information more effectively. However, when using the Transformer to handle time series data, the entire time series is forwarded all at once through the Transformer architecture, making it difficult to extract temporal or sequential dependencies and features [11]. To overcome temporal differences in the Transformer, Kazemi et al. [12] introduced a model-agnostic time vector representation called Time2Vec. Time2Vec captures both periodic and non-periodic features, generating time vector representations. This embedding layer can be incorporated into the Transformer to enhance model performance.

Research has shown that frequency decomposition algorithms can assist deep learning models in better analyzing the decomposed time series [13]. One common approach is to apply one-shot decomposition methods widely used in time series modeling to improve prediction performance. However, the stability of the subseries obtained after one-shot

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

subsequences, with Temporal Convolutional Networks (TCN) then employed for time series forecasting.

Inspired by this, this paper proposes a novel stock prediction model, CEEMD-Time2Vec-Transformer (ETT). This model combines the methods that can capture both periodic and non-periodic features in stock time series, namely CEEMD and Time2Vec, and utilizes the Transformer for prediction.

The remainder of this paper is organized as follows. Section [2](#) provides a review of related work. Section [3](#) explains the proposed methodology. Section [4](#) analyzes the experimental results. The final section concludes with a summary and future prospects.

Our contributions can be summarized as follows:

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

example, Kaczmarek et al. [23] applied the Random Forest (RF) method to predict the constituents of the S&P 500, validate forecast volatility, and analyze the efficiency of portfolio optimization. Yu et al. [24] introduced Principal Component Analysis (PCA) into the SVM model to extract low-dimensional feature information. However, these machine learning model structures may struggle to capture the various feature relationships between sequence heterogeneity [11].

With the development of deep learning, many existing research results have demonstrated that deep learning frameworks outperform traditional machine learning models in financial market prediction. Typical network architectures, such as CNN and RNN, have better predictive capabilities than traditional machine learning algorithms. In paper [25], the LSTM model and Bidirectional LSTM (BiLSTM) model were used with appropriate hyperparameter tuning to predict future stock trends. Deep learning frameworks can learn patterns of temporal variation from historical data, enabling better prediction of future price

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

network based on Transformer Encoder model to extract deep semantic features from social media and then capture the structural relationships of the text using capsule network to predict stock prices. Based on the above research, it can be concluded that Transformer models are one of the accepted deep learning models in the financial area. The aforementioned work demonstrates the successful application of traditional deep learning models in the financial domain. However, traditional neural networks may not adequately learn various effective data feature representations [37], thus reducing the accuracy of analysis.

In paper [14], it was pointed out that combining Transformer with traditional signal analysis algorithms has significant advantages. This is because deep learning models cannot consistently and stably produce high-quality predictions in all scenarios. Utilizing signal analysis methods for time decomposition and feature extraction can enhance the effectiveness of the model. Typical signal decomposition algorithms such as EMD and

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

more effective extraction of pattern information from stock time series, thereby enhancing the accuracy of predictions.

3 Methodology

In this work, we focus on forecasting stock fluctuations. Formally, we use the historical observation sequence of stock data $(\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_T\} \in \mathbb{R}^{T \times F})$, where T is the time steps and F is the number of features. For convenience, we denote $(\mathbf{X}_{t,:})$ as the values recorded at time step t , and $(\mathbf{X}_{:,f})$ as the entire time series of feature f . The model takes the feature matrix $\mathbf{X}$ as input and outputs the predicted label $\mathbf{y}$. The calculation of $\mathbf{y}$ is shown as follows, where (p_T) represents the closing price at time T .

Your privacy, your choice

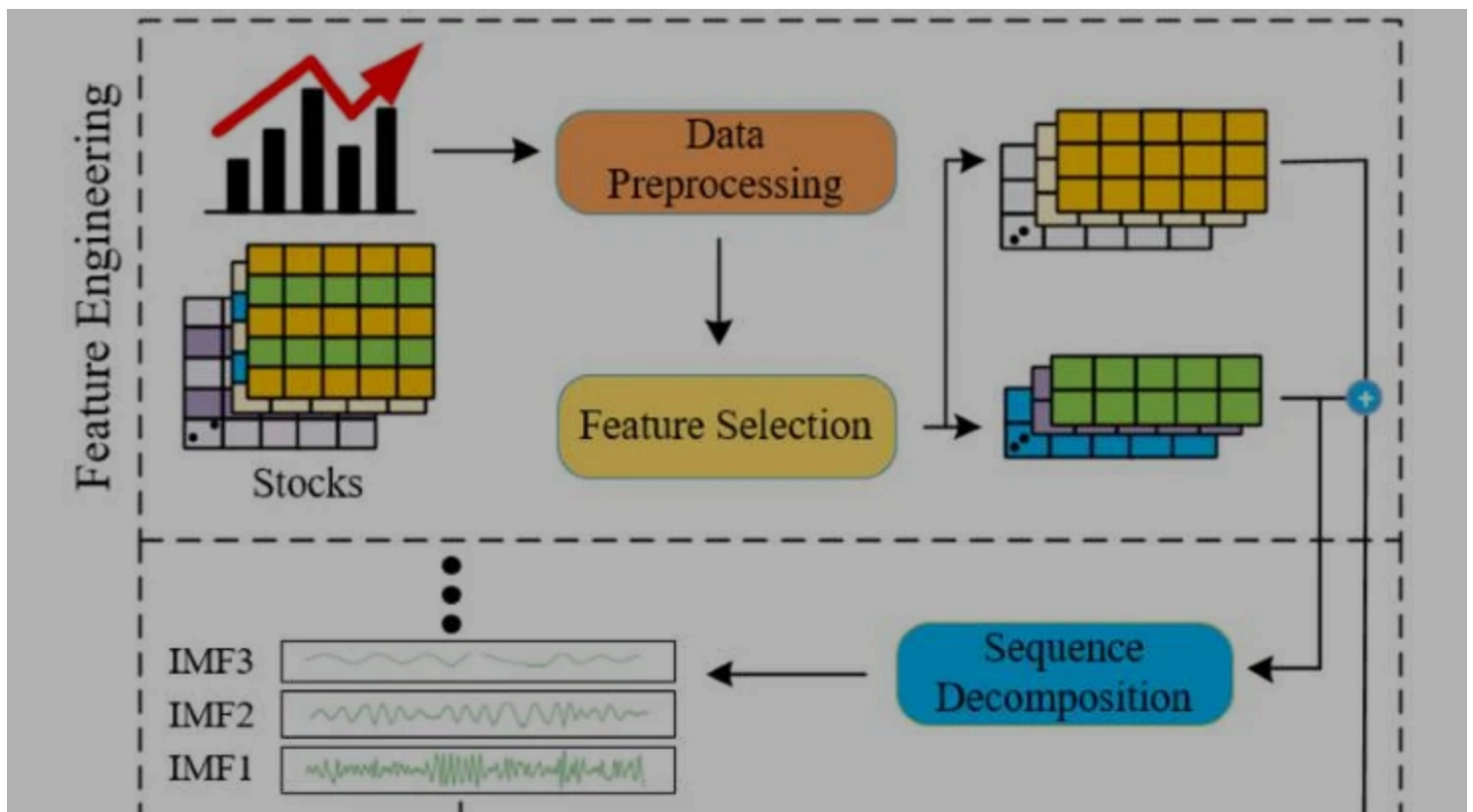
We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**



Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

each feature. Then, we determined the threshold parameter n , commonly set to 3. Finally, we adjusted the feature values that exceeded the range $[(\mu_{\{f\}} - 3\sigma_{\{f\}}), (\mu_{\{f\}} + 3\sigma_{\{f\}})]$ by setting the values exceeding the threshold to the threshold value. The values within the threshold range remained unchanged.

To eliminate the influence of dimensional units among features, the data needs to be standardized. After applying data standardization, the distribution of the original data becomes a standard normal distribution, which is suitable for comprehensive comparative evaluation. The formula for zero-mean normalization is as follows:

$$\mathbf{X}_{:,f}' = \frac{\mathbf{X}_{:,f} - \mu_{\{f\}}}{\sigma_{\{f\}}}$$

(2)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

(2) Add white noise: In each iteration, white noise is added to produce two signals, as shown below:

$$\begin{aligned} \mathbf{P}_i &= \mathbf{X}_{:,f}' + \mathbf{n}_i \\ \mathbf{N}_i &= \mathbf{X}_{:,f}' - \mathbf{n}_i \end{aligned}$$

(3)

(3) EMD decomposition: Perform EMD decomposition on the signals $(\mathbf{P}_i)$ and $(\mathbf{N}_i)$, which include added positive and negative white noise, respectively. This results in q intrinsic mode functions (IMFs), which can be represented as follows:

$$\begin{aligned} \mathbf{P}_i &= \sum_{j=1}^q \mathbf{c}_{j,i}^{(1)} \\ \mathbf{N}_i &= \sum_{j=1}^q \mathbf{c}_{j,i}^{(2)} \end{aligned}$$

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

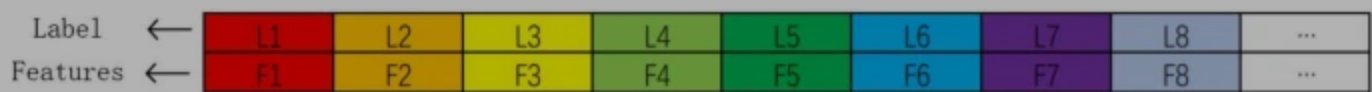
You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

of the stock sequence are decomposed using CEEMD. The decomposed sequence is then combined with the original sequence via residual connections. Figure 2 illustrates the process of constructing features and labels from a single stock, using $(T=3)$ as an example, where $(\mathbf{F}_t)$ represents all features on day t , and $(\mathbf{L}_t)$ represents the label on day t . During the construction process, the first new data point is formed by combining all features and labels from the first three days of the original dataset into a new set of features. These new features are then decomposed using CEEMD, with the label for the fourth day serving as the new label. The subsequent new data points are generated in a similar manner.

Figure 2



Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

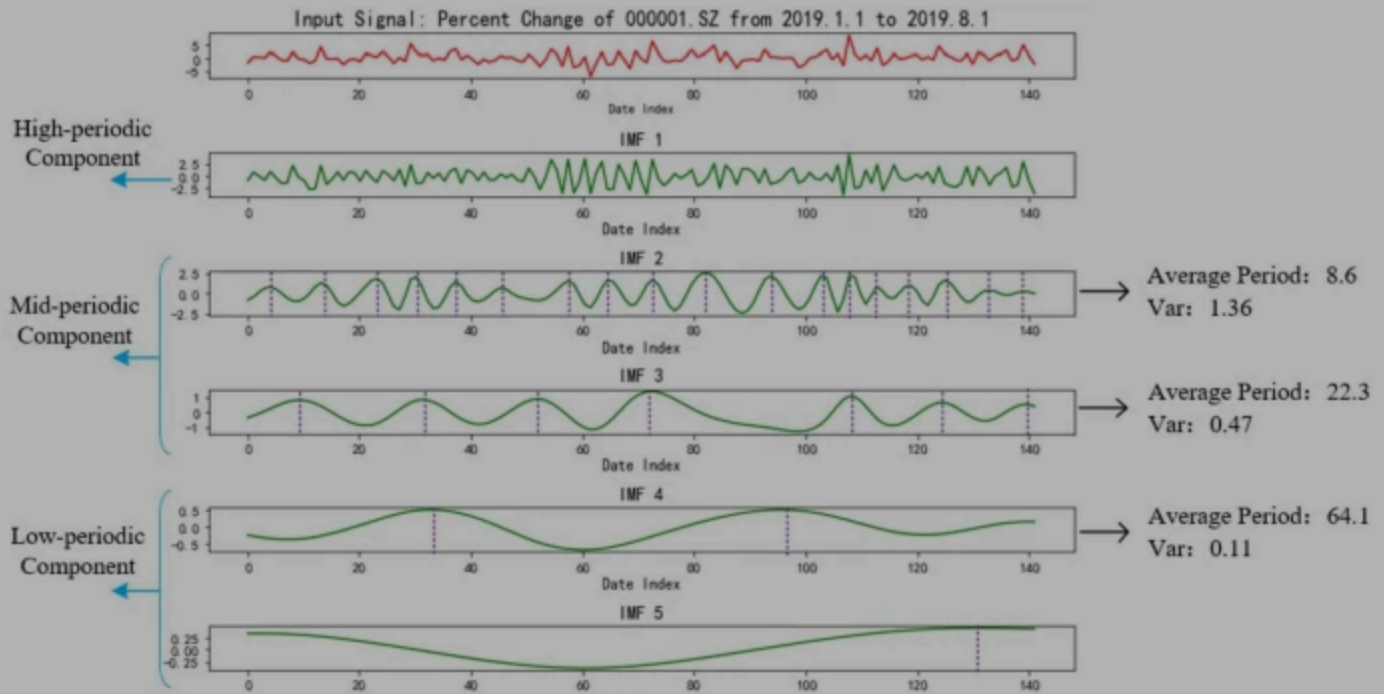
By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Figure 3



Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- **Store and/or access information on a device**
- **Personalised advertising and content, advertising and content measurement, audience research and services development**

Embedding is typically used in natural language processing to map sparse or high-dimensional word vectors to a lower-dimensional space [42]. Similarly, we convert the processed data into embedding representations through the embedding layer. Given the input data $(\{\hat{\mathbf{X}}\})$, the feature sequence generates embedding vectors after passing through the embedding layer. To effectively represent the inherent temporal information in stock time series data, we add positional encoding to the input sequence. Specifically, we use sine and cosine functions of different frequencies to compute these positional encodings, as shown below:

$$\begin{aligned} \mathbf{PE}_{(t,2i)} &= \sin(t/1000^{2i/d_{\text{model}}}) \\ \end{aligned}$$

(7)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

periodic features in stock sequences, thereby enhancing the model's adaptability to different time patterns.

Time2Vec is an orthogonal but complementary approach, which provides a model-agnostic vector representation for time.

$$\begin{aligned} \mathbf{t2v}(\mathbf{H})[i] = \begin{cases} \omega_{\mathbf{H} + \varphi_i}, & 0 \leq i < F(\omega_{\mathbf{H} + \varphi_i}) \\ 1, & \text{else} \end{cases} \end{aligned}$$

(10)

Where k represents the dimension of Time2Vec, F is the periodic activation function, ω

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 partners, also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Taking stock 000001.SZ as an example, Fig. 4 displays a heatmap created by plotting a subset of the parameters $\{\omega_i\}$ learned by Time2Vec. From the figure, it can be observed that the model captures both periodic and non-periodic data. The brightness at $\omega = 0$ is moderate, indicating a lower presence of non-periodic components in the data. The low-frequency portion exhibits brighter colors, suggesting a higher abundance of low-frequency data. In the mid-to-high-frequency range, several dark blocks can be observed from the figure, indicating discontinuities in the data across different periods. Overall, these observations align with the corresponding components in the CEEMD decomposition discussed in the previous section. The captured features are consistent with the components shown in Fig. 3. Therefore, by combining Time2Vec and CEEMD, we can not only effectively filter out noise but also better capture both periodic and non-periodic patterns in stock data, thereby enhancing the predictive performance of the model.

Your privacy, your choice

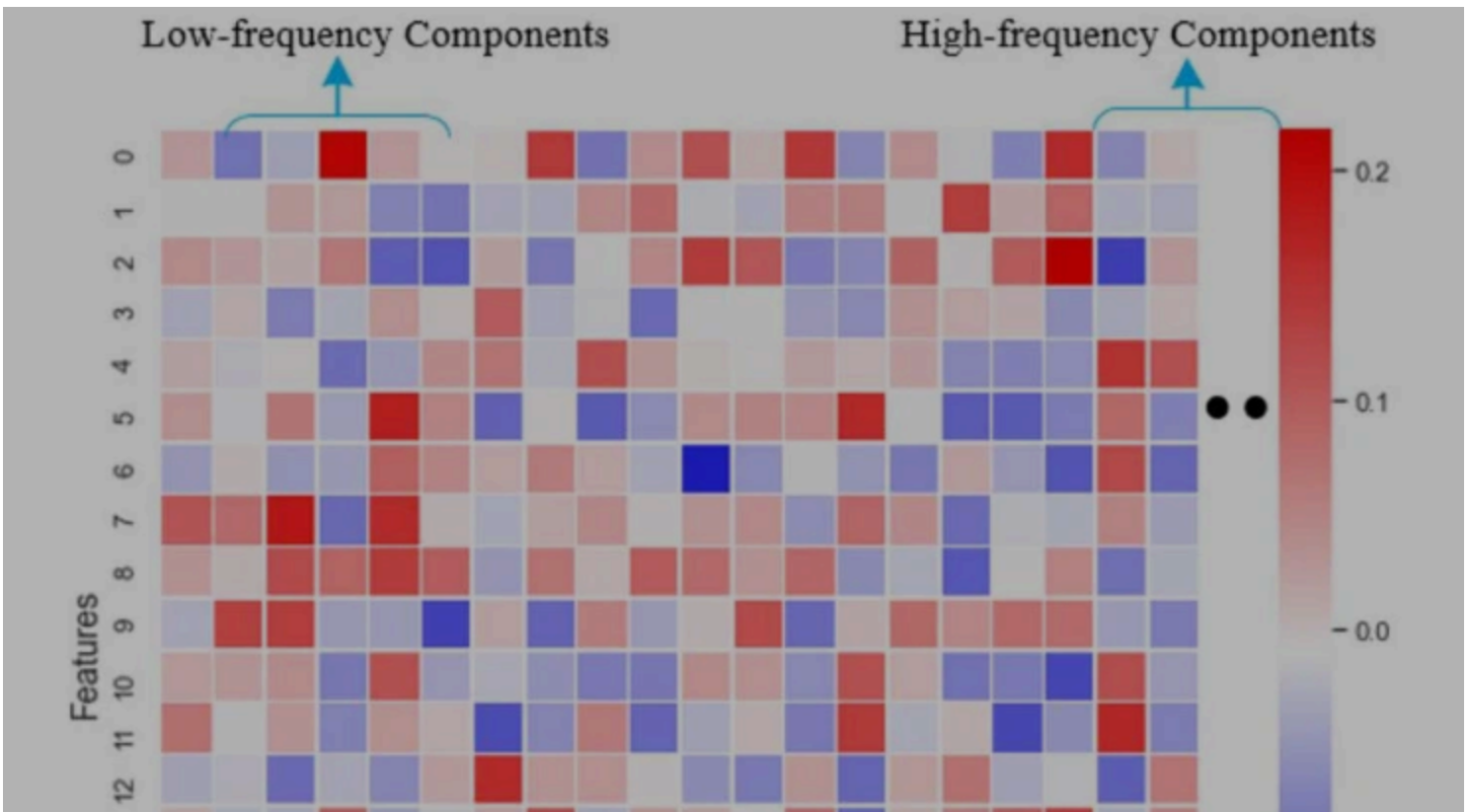
We use essential cookies to make sure the site can function. We, and our 91 partners, also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**



Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

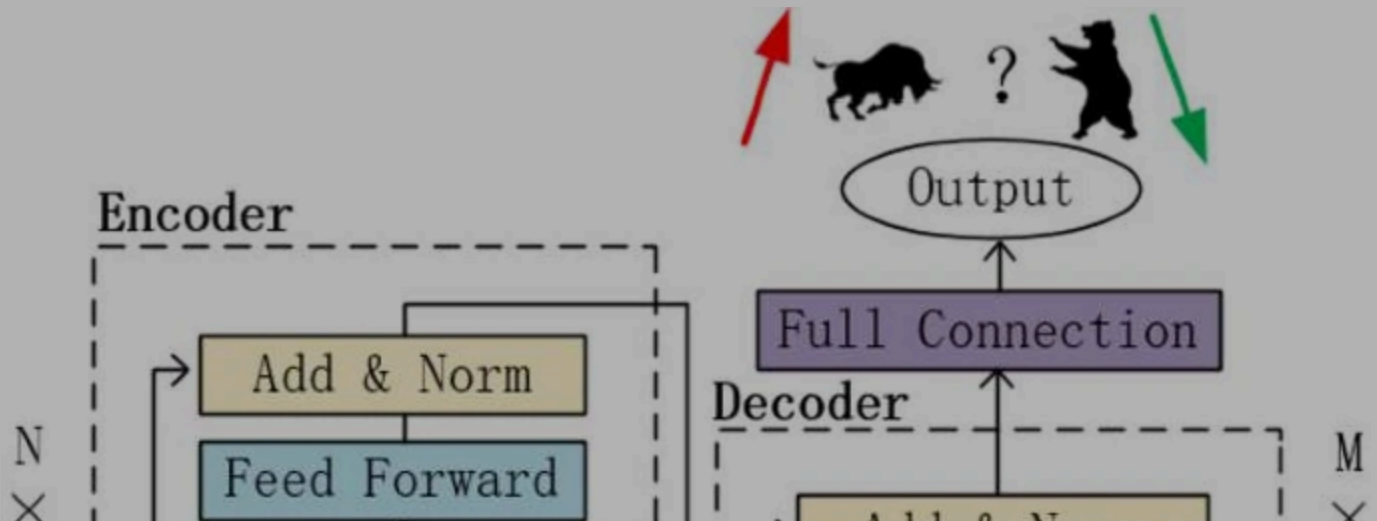
You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

multiple linear layers with leaky ReLU activation functions are used as the decoder of the model. Figure 5 illustrates the architecture of the Transformer used in ETT.

Figure 5



Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

position during the encoding process of a sentence. The hidden vector representation of the entire sentence is then obtained by summing the weighted values. The specific formula is as follows:

$$\mathbf{Q} = \mathbf{Z} \times \mathbf{W}_{\{Q\}}$$

(12)

$$\mathbf{K} = \mathbf{Z} \times \mathbf{W}_{\{K\}}$$

(13)

$$\mathbf{V} = \mathbf{Z} \times \mathbf{W}_{\{V\}}$$

(14)

Your privacy, your choice

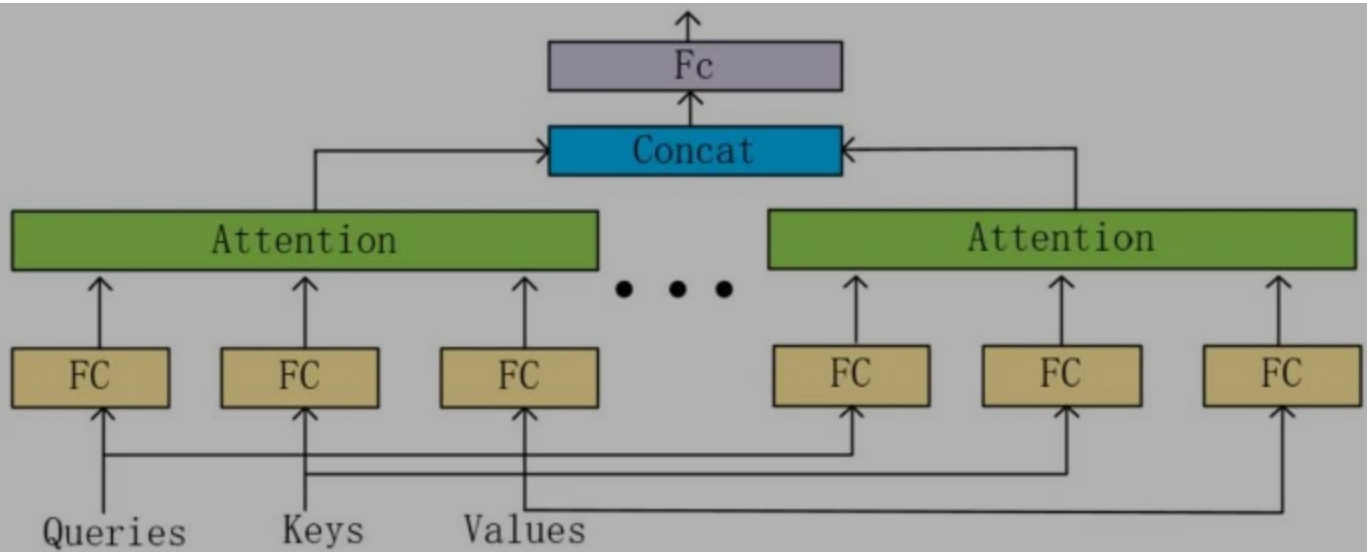
We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**



Architecture of the Multi-Head Attention based Transformer Model

4 Experiments

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

inter-feature correlation. Table 1 presents examples of 10 selected features information. Subsequently, the trade data was right-joined with the Tushare dataset. Outliers were handled for all data, and missing values were filled with the same value as the previous day, followed by normalization.

Table 1 Basic information on ten selected features for stock analysis

4.2 Hyper-parameters setting

To achieve optimal model performance, we employed a grid search strategy during the training process. The specific ranges for these hyperparameters are detailed in Table 2. By thoroughly examining various parameter combinations and independently training models

Your privacy, your choice

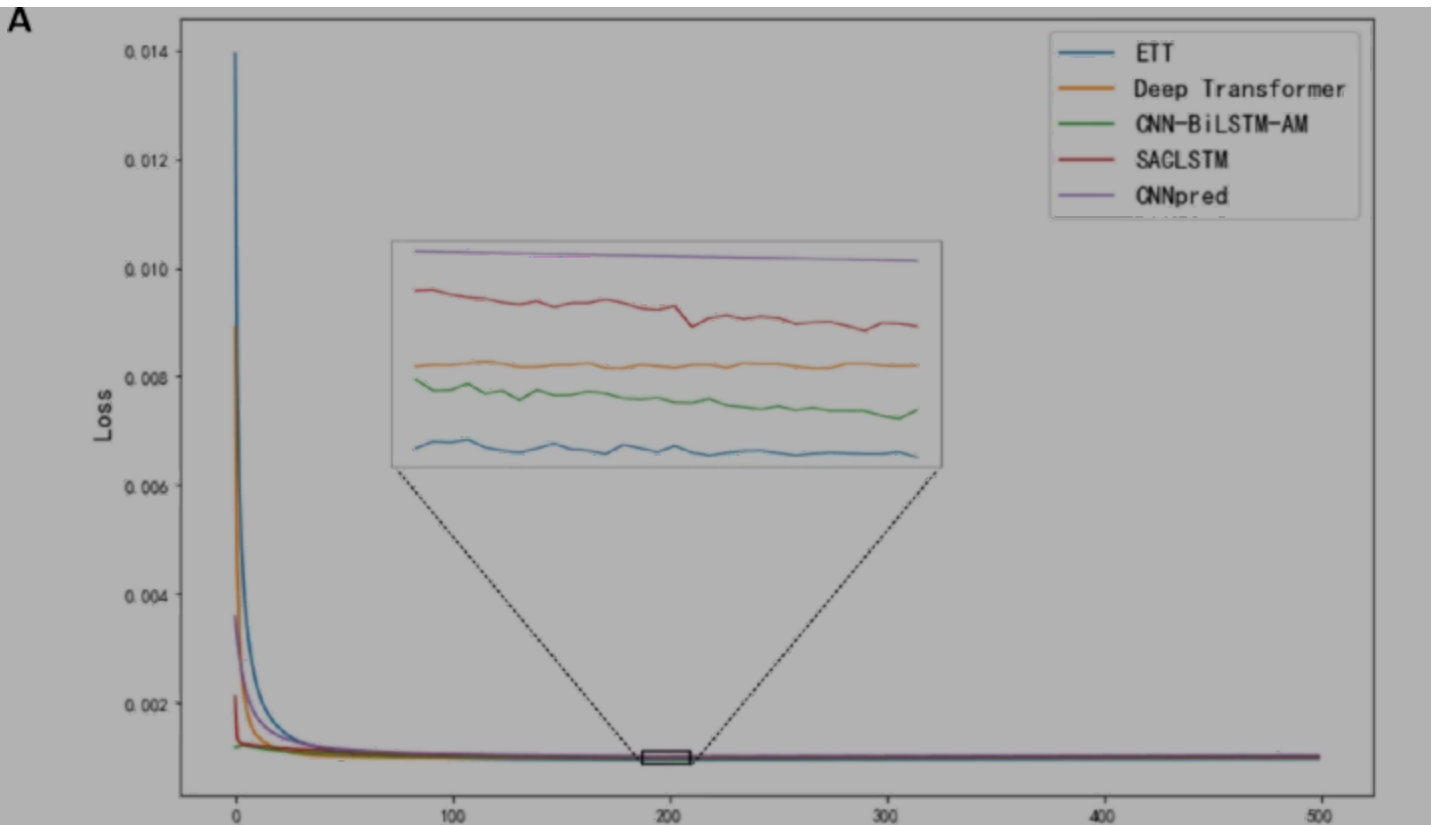
We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**



Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- **Store and/or access information on a device**
- **Personalised advertising and content, advertising and content measurement, audience research and services development**

4.3 Training process and prediction results

Finally, the data is inputted into the adjusted Transformer for training and prediction. In the usage of deep learning models, it is common to evaluate the difference between actual values and predicted values using loss errors. Several commonly used metrics for predicting errors include mean absolute error (MAE), mean squared error (MSE), root mean square error (RMSE), and mean absolute percentage error (MAPE). The calculation formulas for these metrics are as follows:

$$\begin{aligned} \text{MAE} &= \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \end{aligned}$$

(16)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

MAE, MSE, RMSE, and MAPE measure the predictive performance of the model from different perspectives. MAE and MSE quantify the absolute error of predictions, while MAPE assesses the relative error of the model. Smaller values of these metrics indicate better model performance. In this paper, the choice of prediction metric is the percent change, therefore MAE and MSE are selected as evaluation metrics. Table 3 presents the performance of various algorithms on different datasets.

Table 3 Prediction results based of different models on CSI 100 and Hushen 300 datasets. The better results on the validation set are highlighted in *bold* and the second best results are in underlined

From Table 3, it can be observed that the ETT model demonstrates better fitting

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

(2) Select the top n stocks with the highest predicted returns for the day and buy them. The buying weight (w_i) for the i -th stock is determined by its factor exposure to the predicted returns (y_i) .

$$w_i = e^{y_i} / \sum_{j=1}^n e^{y_j}$$

(20)

(3) If it is the last day, calculate the total value of stocks and cash. Otherwise, proceed to the next day.

n is set to 5, and the initial capital is set to 100,000. Transaction fees are not considered. The daily changes in total capital are observed. Figure 8 represents the performance of all algorithms on different datasets. Table 4 provides statistics on total returns, maximum

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

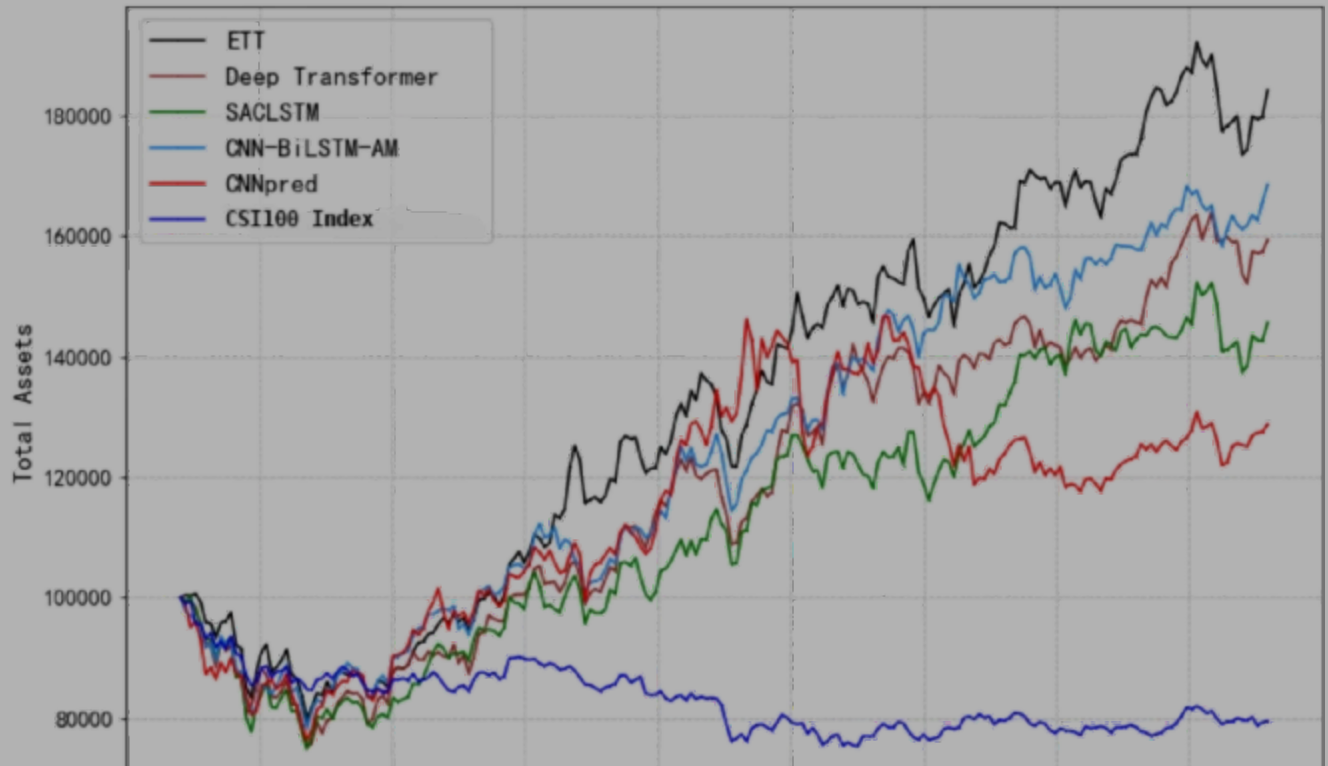
By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

A



Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Table 4 The performance of the trading strategy on CSI 100 and Hushen 300. The better results are highlighted in *bold* and the second best results are in underlined

From Fig. 8 and Table 4, it can be observed that the ETT model achieves the highest cumulative returns for investors in the CSI 100 and Hushen 300 datasets. Although the CNNpred model sometimes accurately identifies stocks with high short-term returns, its poor robustness results in capturing stocks that decline, leading to the lowest cumulative returns. Additionally, the stock sequence array convolutional LSTM (SACLSTM) and CNN-BiLSTM-AM models exhibit relatively stable increases in total investment assets, but their

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

investors with more effective and reliable references. The effectiveness of each module of the model is further demonstrated through ablation studies conducted in this research.

However, there are some limitations to this study. It only considers the prediction of two Chinese stock datasets. The correlation between stock markets in many countries is high. Therefore, it would be meaningful to model and experiment with stock data from various countries worldwide. Additionally, exploring the application of the ETT model in other financial markets such as futures markets and bond markets would also be a valuable area for further exploration.

Availability of data and material

The datasets used or analysed during the current study are available from the corresponding

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

CNN: Convolutional Neural Networks

RNN: Recurrent Neural Networks

LSTM: Long Short-Term Memory Networks

BiLSTM: Bidirectional LSTM

IMF: Intrinsic Mode Function

RF: Random Forest

PCA: Principal Component Analysis

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

MAE: Mean Absolute Error

RMSE: root mean square error

MAPE: Mean Absolute Percentage error

SACLSTM: Stock Sequence Array Convolutional LSTM

AM: Attention Model

References

1. Kohara K, Fukuhara Y, Nakamura Y (1996) Selective presentation learning for neural

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

5. Wilhelmsson A (2006) Garch forecasting performance under different distribution assumptions. J Forecast 25:561–578

6. Lorenzato de Oliveira JF, Ludermir TB (2014) A distributed pso-arima-svr hybrid system for time series forecasting. In: 2014 IEEE international conference on Systems, Man, and Cybernetics (SMC), pp 3867–3872

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

11. Zhou F, Zhang Q, Zhu Y, Li T (2023) T2v_tf: an adaptive timing encoding mechanism based transformer with multi-source heterogeneous information fusion for portfolio management: a case of the Chinese a50 stocks. Expert Syst Appl 213:119020
[Article](#) [Google Scholar](#)
12. Kazemi SM, Goel R, Eghbali S, Ramanan J, Sahota J, Thakur S, Wu S, Smyth C, Poupart P, Brubaker M (2019) Time2vec: Learning a vector representation of time. arXiv preprint. arXiv:[1907.05321](#)
13. Wang K, Qi X, Liu H, Song J (2018) Deep belief network based k-means cluster approach for short-term wind power forecasting. Energy 165:840–852
[Article](#) [MATH](#) [Google Scholar](#)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

17. Huang NE, Shen Z, Long SR, Wu MC, Shih HH, Zheng Q, Yen N-C, Tung CC, Liu HH (1998) The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis. Proc R Soc Lond, Ser A, Math Phys Eng Sci 454(1971):903–995

[Article](#) [MathSciNet](#) [MATH](#) [Google Scholar](#)

18. Siarni-Namini S, Tavakoli N, Namin AS (2019) The performance of lstm and bilstm in forecasting time series. In: 2019 IEEE international conference on big data (big data). IEEE, pp 3285–3292

[Chapter](#) [MATH](#) [Google Scholar](#)

19. Chen L, Chi Y, Guan Y, Fan J (2019) A hybrid attention-based emd-lstm model for

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

22. Long W, Lu Z, Cui L (2019) Deep learning-based feature engineering for stock price movement prediction. Knowl-Based Syst 164:163–173

[Article](#) [MATH](#) [Google Scholar](#)

23. Kaczmarek T, Perez K (2022) Building portfolios based on machine learning predictions. Econ Res 35(1):19–37

[MATH](#) [Google Scholar](#)

24. Yu H, Chen R, Zhang G (2014) A svm stock selection model within pca. Proc Comput Sci 31:406–412

[Article](#) [MATH](#) [Google Scholar](#)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

28. Sirignano J, Cont R (2019) Universal features of price formation in financial markets: perspectives from deep learning. Quant Finance 19(9):1449–1459

[Article](#) [MathSciNet](#) [MATH](#) [Google Scholar](#)

29. Kaushik M, Giri A (2020) Forecasting foreign exchange rate: a multivariate comparative analysis between traditional econometric, contemporary machine learning & deep learning techniques. arXiv preprint. arXiv:[2002.10247](#)

30. Selvin S, Vinayakumar R, Gopalakrishnan E, Menon VK, Soman K (2017) Stock price prediction using lstm, rnn and cnn-sliding window model. In: 2017 International Conference on Advances in Computing, Communications and Informatics (icacci). IEEE, pp 1643–1647

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

34. Wang C, Chen Y, Zhang S, Zhang Q (2022) Stock market index prediction using deep transformer model. Expert Syst Appl 208:118128

[Article](#) [MATH](#) [Google Scholar](#)

35. Ding Q, Wu S, Sun H, Guo J, Guo J (2020) Hierarchical multi-scale Gaussian transformer for stock movement prediction. In: IJCAI, pp 4640–4646

[Google Scholar](#)

36. Malibari N, Katib I, Mehmood R (2021) Predicting stock closing prices in emerging markets with transformer neural networks: the saudi stock exchange case. Int J Adv Comput Sci Appl 12(12)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

> **Store and/or access information on a device**

> **Personalised advertising and content, advertising and content measurement, audience research and services development**

41. Torres ME, Colominas MA, Schlotthauer G, Flandrin P (2011) A complete ensemble empirical mode decomposition with adaptive noise. In: 2011 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP). IEEE, pp 4144–4147

[Chapter](#) [MATH](#) [Google Scholar](#)

42. Mikolov T, Chen K, Corrado G, Dean J (2013) Efficient estimation of word representations in vector space. arXiv preprint. arXiv:[1301.3781](#)
43. Tallec C, Ollivier Y (2018) Can recurrent neural networks warp time? arXiv preprint. arXiv:[1804.11188](#)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

48. Giacomini R, White H (2006) Tests of conditional predictive ability. *Econometrica* 74(6):1545–1578

[Article](#) [MathSciNet](#) [MATH](#) [Google Scholar](#)

49. Hoseinzade E, Haratizadeh S (2019) Cnnpred: cnn-based stock market prediction using a diverse set of variables. *Expert Syst Appl* 129:273–285

[Article](#) [MATH](#) [Google Scholar](#)

50. Wu JM-T, Li Z, Herencsar N, Vo B, Lin JC-W (2021) A graph-based cnn-lstm stock price prediction algorithm with leading indicators. *Multimed Syst*:1–20

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Authors and Affiliations

School of Economics, Zhejiang University of Technology, Hangzhou, China

Cheng Zhao

School of Computer Science, Zhejiang University of Technology, Hangzhou, China

Junyi Cai & Shuyi Yang

Contributions

ZC proposed the central idea of the paper, drafted the manuscript, and led the overall design of the study. CJ conducted the data analysis, contributed to the interpretation of the results, and revised the manuscript. YS wrote the code and implemented the computational models. All authors read and approved the final manuscript.

Corresponding author

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to [springer.com](#) and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Open Access This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

[Reprints and permissions](#)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 [partners](#), also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**

Anyone you share the following link with will be able to read this content:

[Get shareable link](#)

Provided by the Springer Nature SharedIt content-sharing initiative

Keywords

[CEEMD](#)

[Time2Vec](#)

[Transformer](#)

[Periodic/Non-Periodic Features](#)

Your privacy, your choice

We use essential cookies to make sure the site can function. We, and our 91 , also use optional cookies and similar technologies for advertising, personalisation of content, usage analysis, and social media.

By accepting optional cookies, you consent to allowing us and our partners to store and access personal data on your device, such as browsing behaviour and unique identifiers. Some third parties are outside of the European Economic Area, with varying standards of data protection. See our [privacy policy](#) for more information on the use of your personal data. Your consent choices apply to springer.com and applicable subdomains.

You can find further information, and change your preferences via 'Manage preferences'. You can also change your preferences or withdraw consent at any time via 'Your privacy choices', found in the footer of every page.

We use cookies and similar technologies for the following purposes:

- > **Store and/or access information on a device**
- > **Personalised advertising and content, advertising and content measurement, audience research and services development**